

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 7 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTECOE705B - CYBER SECURITY & BLOCK CHAIN TECHNOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The process of converting plaintext into ciphertext using the same key for both encryption and decryption is called _____.
 - a) Symmetric key cryptography
 - b) Asymmetric key cryptography
 - c) Hashing
 - d) Steganography
- 2 Match the following cryptographic techniques with their correct types:

A. RSA	1. Asymmetric Key
B. DES	2. Classical
C. Vigenère Cipher	3. Block Cipher
D. Elliptic Curve	4. Modern Asymmetric

 - a) A-1, B-3, C-2, D-4
 - b) A-3, B-4, C-2, D-1
 - c) A-2, B-1, C-3, D-4
 - d) A-4, B-3, C-1, D-2
- 3 Which of the following ensures non-repudiation in digital communication?
 - a) MAC
 - b) Hash function
 - c) Digital signature
 - d) Pseudorandom function
- 4 Which cipher type encrypts one bit or byte at a time rather than blocks of data?
 - a) Block cipher
 - b) Polyalphabetic cipher
 - c) Stream cipher
 - d) Homomorphic cipher
- 5
 - A. RSA
 - B. AES
 - C. SHA-256
 - D. HMAC
 1. Block cipher
 2. Public key encryption
 3. Message hashing
 4. Authentication code
 - a) A-3, B-2, C-1, D-4
 - b) A-2, B-1, C-3, D-4
 - c) A-1, B-3, C-4, D-2
 - d) A-4, B-2, C-3, D-1
- 6 The _____ algorithm is used in blockchain to solve proof-of-work puzzles.
 - a) RC4
 - b) SHA-256
 - c) RSA
 - d) ElGamal
- 7 Which cryptographic algorithm is based on the difficulty of solving discrete logarithms on elliptic curves?
 - a) DES
 - b) ElGamal
 - c) ECC
 - d) RSA
- 8 Firewalls are used in network security to _____.
 - a) Encrypt traffic
 - b) Detect spam
 - c) Filter traffic
 - d) Manage keys

- 9 Match the following cryptographic properties:
- | | |
|--------------------|--------------------------------------|
| A. Integrity | 1. Prevent tampering |
| B. Confidentiality | 2. Only authorized users access data |
| C. Non-repudiation | 3. Prove sender's identity |
| D. Authentication | 4. Confirm identity |
- a) A-1, B-2, C-3, D-4 b) A-2, B-1, C-4, D-3
c) A-3, B-4, C-1, D-2 d) A-4, B-3, C-2, D-1
- 10 The _____ algorithm is commonly used in blockchain for generating digital signatures.
a) SHA-1 b) MD5 c) DSA d) RSA
- 11 Which of the following is a primary goal of cryptography?
a) Compression b) Confidentiality c) Storage d) Multiplexing
- 12 The main objective of elliptic curve cryptography is to provide high security with _____ key sizes.
a) Small b) Large c) Random d) Constant

Q2. Solve the following.

- A) Using the Euclidean algorithm, find the greatest common divisor of the pair of integers 88 and 220. 6
- B) Define modern cryptography and explain its primary objectives. List and describe the key differences between symmetric and asymmetric cryptosystems. 6

Q3. Solve the following.

- A) Explain the core functions of an S-Box in block ciphers. Solve: Encrypt the plaintext "11010111" using one round of a simplified AES with a given key and S-Box. 6
- B) Explain the structure and working of DES with a neat diagram. Compare the structure and security features of DES and AES. 6

Q4. Solve Any Two of the following.

- A) Define stream cipher and give two examples. Given a binary stream of plaintext and a keystream:
Plaintext: 10110100
Keystream: 11001010
Encrypt using a stream cipher and show the result. 6
- B) List the steps of the RSA algorithm. Using RSA, Compute n , $\phi(n)$, and the private key d for the following values: $p = 13$, $q = 17$, and $e = 5$. 6
- C) Explain the Merkle–Damgård construction and its core idea. Summarize the structure of the Merkle–Damgård construction with a diagram. 6

Q5. Solve Any Two of the following.

- A) Define a digital signature. List its key properties. Compute the digital signature for a message hash $h = 91$, using RSA parameters: $p = 11$, $q = 13$, $e = 7$. Show the signature and verification. 6
- B) What are the basic parameters used in Elliptic Curve Cryptography (ECC)? Explain how elliptic curves are used in key generation with a suitable example. 6

- C) Describe the difference between blind signatures and undeniable signatures. 6
Differentiate between RSA-based digital signatures and DSA-based signatures.

Q6. Solve Any Two of the following.

- A) Explain the concept of immutability and its importance in distributed ledgers. 6
Describe the difference between deterministic and probabilistic consensus.
- B) Define blockchain and state any three of its key characteristics. Design a simple 6
blockchain structure with 3 blocks and show how transactions, hashes, and nonces
are used.
- C) Explain how Merkle Trees are used in blockchain to verify transactions. Describe 6
how blockchain achieves trust in a trustless environment.

*** End ***